GEOTECHNICAL ENGINEERING

UNIT - I : INTRODUCTION & INDEX PROPERTIES OF SOILS

Course Code: 23A01503T

Syllabus: Soil formation - Soil structure - Adsorbed water - Mass-volume relationship - Relative density. Index properties: moisture content, specific gravity, in-situ density, grain size analysis (sieve and hydrometer methods), consistency limits and indices - I.S. classification of soils.

1. Introduction

Geotechnical engineering is the branch of civil engineering that deals with the behaviour of earth materials (soil and rock) and their use as foundation support or construction material.

- Soil = unconsolidated, loose or weakly bonded accumulation of mineral particles formed by weathering of rocks, with voids containing air and/or water.
- Rock = hard, naturally formed, firmly bonded mass of mineral matter.
- Soil mechanics = application of the laws of mechanics and hydraulics to soils (Terzaghi - "Father of Soil Mechanics", 1925, Erdbaumechanik).
- Soil is a three-phase system: solids + water + air.

Why study soil?

- Soil is the foundation for every structure: buildings, bridges, dams, roads, runways.
- Soil is the construction material: embankments, earth dams, road subgrade.
- Soil is retained / supported: retaining walls, slopes, excavations, tunnels.

Main areas of Geotechnical Engineering

- Site investigation and soil exploration
- Foundation engineering (shallow and deep foundations)
- Earth pressure and retaining structures, slope stability
- Earth dams, embankments, pavements (subgrade design)
- Ground improvement (compaction, grouting, soil stabilisation, geosynthetics)

2. Soil Formation

Soils are formed by the weathering (disintegration and decomposition) of parent rock by the action of natural agents like temperature, water, wind, ice and living organisms.

(a) Physical (mechanical) weathering - disintegration

- Changes in temperature: day-night expansion and contraction cause cracks and spalling.
- Frost action: water in cracks freezes, expands by about 9% and wedges the rock apart.
- Abrasion by wind, running water, glaciers and wave action.
- Plant roots and burrowing animals widen the cracks.
  - Result: coarse-grained soils - gravel, sand, silt (bulk minerals unchanged).

(b) Chemical weathering - decomposition

- Hydration, hydrolysis, oxidation, carbonation, solution and leaching change the minerals chemically.
- Feldspar + water + CO_2 gives clay minerals (kaolinite, illite, montmorillonite).
  - Result: fine-grained soils - clays (very small particles, large surface area).

(c) Biological weathering

- Lichens, mosses, bacteria, roots and burrowing animals produce acids and organic matter (humus).

Physical weathering produces silt, sand and gravel, while chemical weathering produces clay minerals. Rate of chemical weathering is high in hot and humid climates.

Classification of soils based on origin

Soil type	Transported by	Features / examples
Residual soil	not transported (formed in place)	Grading from fine at top to rock at depth, e.g. laterite, black cotton soil
Alluvial soil	running water (rivers)	Stratified, sand-silt-clay in layers, fertile; Indo-Gangetic plain
Lacustrine	lake deposits	Fine silt and clay, soft, compressible
Marine soil	sea water	Soft marine clays, high salt content, sensitive
Aeolian soil	wind	Loess, dune sand: uniform, fine, loosely packed
Glacial soil	glaciers / ice	Boulder clay (till), very mixed sizes
Colluvial soil	gravity	Talus / landslide debris at foot of slopes

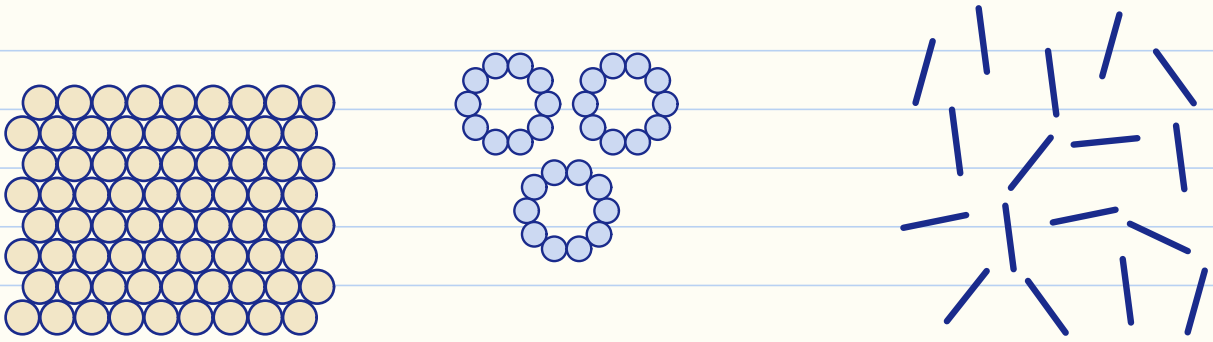
Some Indian soils

- Black cotton soil: montmorillonite, high swelling and shrinkage, Deccan plateau, poor for foundations.

- Laterite soil: rich in iron and aluminium oxides, red, used as road material.
- Alluvial soil: Indo-Gangetic plains and river deltas.
- Desert (aeolian) soil: Rajasthan, very fine uniform sand.

3. Soil Structure

The structure of a soil is the geometrical arrangement of its particles and the forces between them. It governs strength, permeability and compressibility.



(a) Single grained (b) Honeycomb (c) Flocculent

Types of structure

- Single-grained: coarse particles (sand, gravel) settle under gravity; each grain touches its neighbours. Dense or loose packing. Particle weight >> surface forces.
- Honeycomb: fine sands / silts (0.02 - 0.002 mm) settling in water bridge over to form arches around voids. Open structure, high void ratio, can carry static load but collapses under shock.
- Flocculent: clay particles (< 0.002 mm) in salt water. Edge (+ve) of one plate attracts face (-ve) of another giving a "card-house" structure - very high void ratio and strength.
- Dispersed: clay particles in fresh water repel each other, lie parallel to one another - low permeability, low strength after remoulding.
- Coarse-grained skeleton: large grains in contact with fines filling the voids between them.
- Cohesive matrix: coarse grains floating in a matrix of clay.

Flocculated structure has higher strength but is sensitive - when remoulded (disturbed) it breaks down and strength drops (sensitivity). Dispersed structure has lower permeability.

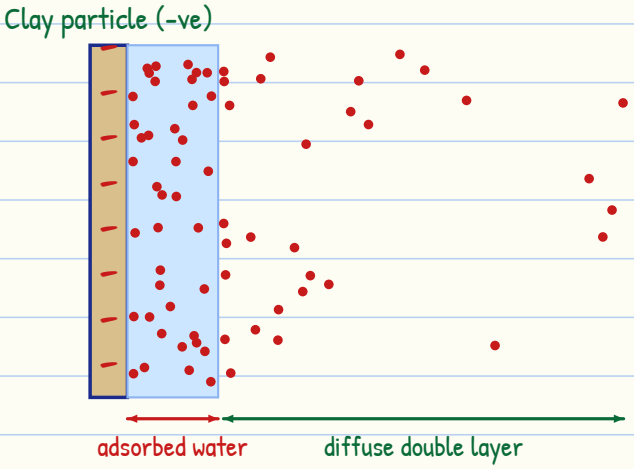
4. Clay Minerals

Clay minerals are hydrous aluminium silicates built from two basic units: the silica tetrahedron (Si) and the alumina octahedron (Al, gibbsite). Sheets stack to form layers.

Mineral	Structure	Bonding	Characteristics
Kaolinite	1:1 (Si + Al sheet)	H-bond (strong)	stable, low swelling, low plasticity
Illite	2:1, K ions between layers	K-bond (medium)	medium swelling and plasticity
Montmorillonite	2:1 (Al between 2 Si)	van der Waals (weak)	very high swelling and plasticity

- Isomorphous substitution: replacement of one cation by another of lower valency (e.g. Al^{3+} by Mg^{2+}) without changing crystal shape gives clay particles a net negative charge.
- Specific surface (surface area per unit mass): sand about 0.0001 m^2/g, kaolinite 10 - 20, illite 65 - 100, montmorillonite 700 - 840 m^2/g.
- Activity of clay = PI / (% clay fraction < 2 micron): inactive < 0.75, normal 0.75 - 1.25, active > 1.25 (montmorillonite about 7).

5. Adsorbed Water



+ve cations (red dots) are attracted to the negative clay surface; ion density falls with distance from the surface.

Water molecules are dipoles - they are held tightly near the surface (adsorbed water: very viscous, behaves like a solid).

Free / pore water lies beyond.

- Clay particles carry a negative charge on their surfaces and positive charge on their edges.
- Water molecules are dipoles (H^+ and O^{2-}); they are attracted to the clay surface by hydrogen bonds, by cations that are themselves hydrated, and by dipole attraction.
- Water held on the surface of the particle by these electro-chemical forces is called adsorbed water (also hygroscopic / bound water). It cannot be removed by gravity, and

extends to about 0.005 micron.

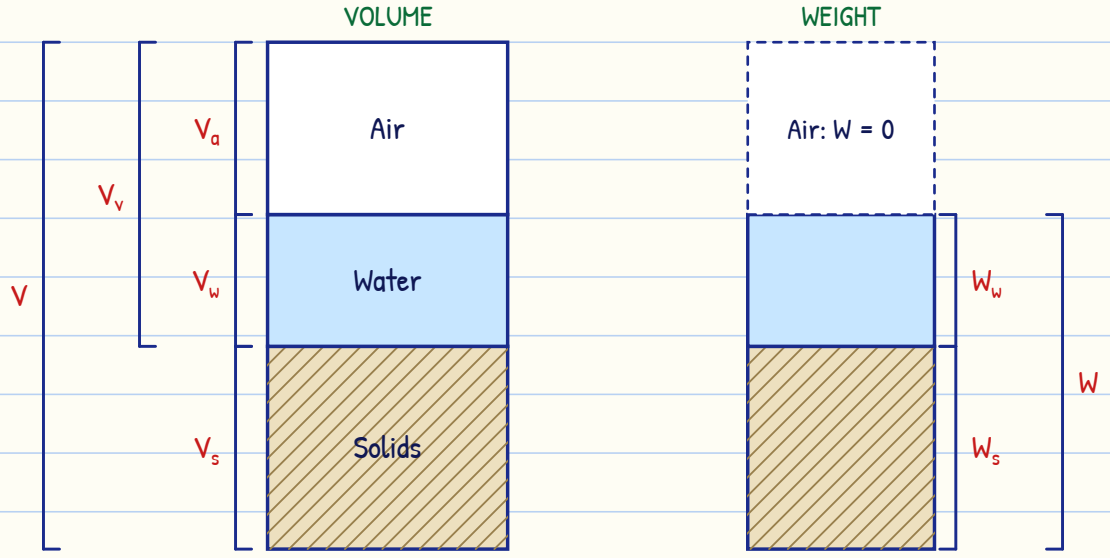
- The layer of adsorbed water + the cations around the particle is called the diffuse double layer. Thickness depends on valency, concentration of the electrolyte, and temperature.
- Thickness reduces when cation valency or concentration increases (e.g. Na^{+} gives thick, Ca^{2+} / Al^{3+} gives thin layer).
- Adsorbed water is much more viscous and has density a little higher than free water; it behaves like a solid and is removed only by oven drying at 105 - 110 degree C.

Effects of adsorbed water

- Gives plasticity and cohesion to clays (more adsorbed water, more plastic).
- Controls consistency limits, swelling, compressibility and permeability of clay.
- Makes clay structure flocculated or dispersed depending upon the double layer thickness.

6. Mass - Volume Relationships

Soil = solids + water + air. For analysis the three phases are shown separately in a phase diagram (three-phase diagram).



- V = total volume, V_s = volume of solids, V_w = volume of water, V_a = volume of air, V_v = $V_w + V_a$ = volume of voids.
- W = total weight, W_s = weight of solids, W_w = weight of water; weight of air is taken as zero. So $W = W_s + W_w$.

Volumetric ratios

- Void ratio $e = V_v / V_s$ (no unit; can be more than 1; sand 0.4 – 1.0, clay 0.5 – 1.5+).
- Porosity $n = V_v / V$ (expressed in %, always less than 100%).
- Degree of saturation $S = V_w / V_v$ (0 for dry soil, 100% for saturated soil).
- Air content $a_c = V_a / V_v$; so $a_c = 1 - S$.
- Percentage air voids $n_a = V_a / V$.

Weight ratios

- Water (moisture) content $w = W_w / W_s$ (can exceed 100%, e.g. peat, soft clays).
- Specific gravity of solids $G = (W_s/V_s) / \gamma_w$ (also called true or grain specific gravity).
- Mass specific gravity $G_m = \gamma / \gamma_w$ (bulk or apparent specific gravity).

Unit weights (densities)

Name	Symbol	Definition
Bulk (total) unit weight	γ	W / V
Dry unit weight	γ_d	W_s / V
Saturated unit weight	γ_{sat}	W / V when $S = 1$
Submerged (buoyant) unit weight	γ'	$\gamma_{sat} - \gamma_w$
Unit weight of solids	γ_s	$W_s / V_s = G \gamma_w$
Unit weight of water	γ_w	9.81 kN/m^3 (1 g/cc or 1000 kg/m^3)

Derivation of important relations (take $V_s = 1$, so $V_v = e$ and $V = 1 + e$)

- Relation between n and e: $n = V_v/V = e/(1+e)$ and $e = n/(1-n)$
- $W_s = G \gamma_w V_s = G \gamma_w$ and $W_w = w W_s = w G \gamma_w$
- $V_w = W_w/\gamma_w = w G$. But $S = V_w/V_v = wG/e$.

$$S e = w G$$

- Dry unit weight $\gamma_d = W_s/V = G \gamma_w / (1 + e)$
- Bulk unit weight $\gamma = (W_s + W_w)/V = (1 + w) G \gamma_w / (1 + e) = (G + S e) \gamma_w / (1 + e)$

$$\gamma_d = \gamma / (1 + w)$$

- Saturated ($S = 1$): $\gamma_{sat} = (G + e) \gamma_w / (1 + e)$
- Submerged: $\gamma' = \gamma_{sat} - \gamma_w = (G - 1) \gamma_w / (1 + e)$
- Dry soil ($S = 0$): $\gamma = \gamma_d = G \gamma_w / (1 + e)$
- Air voids: $n_a = n (1 - S) = (1 - S) e / (1+e)$
- Zero air voids line ($S = 1$): $\gamma_d = G \gamma_w / (1 + w G)$

Order of unit weights: $\gamma_{sat} > \gamma > \gamma_d > \gamma'$. Also $e = (G \gamma_w / \gamma_d) - 1$.

Solved example 1

A soil has bulk density 1.92 g/cc, water content 12% and G = 2.70. Find dry density, void ratio, porosity and degree of saturation.

- $\gamma_d = \gamma/(1 + w) = 1.92/1.12 = 1.714$ g/cc
- $e = G\gamma_w/\gamma_d - 1 = 2.70/1.714 - 1 = 0.575$
- $n = e/(1+e) = 0.575/1.575 = 0.365 = 36.5$ %
- $S = wG/e = (0.12 \times 2.70)/0.575 = 0.563 = 56.3$ %

Solved example 2

A fully saturated clay has w = 45% and G = 2.72. Find e, γ_{sat} and γ'.

- S = 1 so $e = wG = 0.45 \times 2.72 = 1.224$
- $\gamma_{sat} = (G + e)\gamma_w/(1 + e) = (2.72 + 1.224) \times 9.81/2.224 = 17.40$ kN/m^3
- $\gamma' = 17.40 - 9.81 = 7.59$ kN/m^3

7. Relative Density (Density Index)

Used for cohesionless soils to say how dense a sand is in its natural state compared with its loosest and densest states.

$$D_r = (e_{max} - e) / (e_{max} - e_{min}) \times 100\ \%$$

- Also $D_r = [\ \gamma_{d\,max}(\gamma_d - \gamma_{d\,min})\ ] / [\ \gamma_d(\gamma_{d\,max} - \gamma_{d\,min})\ ] \times 100$
- e_{max} = void ratio in loosest state (poured through funnel), e_{min} = void ratio in densest state (vibration + surcharge), e = in-situ void ratio.
- D_r = 0 for the loosest and D_r = 100 % for the densest state.

Relative density (%)	0 - 15	15 - 35	35 - 65	65 - 85	85 - 100
Description	Very loose	Loose	Medium dense	Dense	Very dense

Example

Sand: $e_{max} = 0.85$, $e_{min} = 0.40$, natural e = 0.62.

$D_r = (0.85 - 0.62)/(0.85 - 0.40) = 0.23/0.45 = 0.511 = 51.1$ % - medium dense sand.

Relative density is related to Standard Penetration Test N-value: N < 4 very loose, 4-10 loose, 10-30 medium, 30-50 dense, > 50 very dense. (Terzaghi and Peck)

8. Index Properties of Soils

Index properties are simple tests on disturbed samples used to identify and classify soils and to correlate their engineering behaviour (strength, compressibility, permeability). Two types: soil grain properties (grain size, shape, specific gravity) and soil aggregate properties (water content, consistency, density, relative density).

8.1 Moisture (water) content - w

$$w = W_w / W_s \times 100 = (W_2 - W_3) / (W_3 - W_1) \times 100$$

- Oven drying method (IS 2720 Part 2): W_1 = wt of empty container, W_2 = container + wet soil, W_3 = container + oven dry soil. Dry at 105 - 110 degree C for 24 hours (till constant mass). Most accurate, standard method.
- Pycnometer method: quick, for coarse-grained soil; $w = [(W_2 - W_1) / (W_3 - W_4) \times (G - 1)/G - 1] \times 100$.
- Calcium carbide (Speedy moisture meter): soil + CaC_2 reacts with water giving acetylene gas, whose pressure on the gauge gives moisture. Field use, 5 min.
- Others: sand bath, alcohol method, infrared lamp, torsion balance, radio-isotope method.

8.2 Specific gravity of soil solids - G

$$G = (W_2 - W_1) / [(W_2 - W_1) - (W_3 - W_4)]$$

- Density bottle / pycnometer method (IS 2720 Part 3): W_1 = empty bottle, W_2 = bottle + dry soil, W_3 = bottle + soil + water, W_4 = bottle + water only.
- Procedure: weigh the dry bottle, add 200 g (or 10 - 25 g in a density bottle) of oven-dry soil, add water, remove entrapped air by vacuum / boiling, fill to mark, weigh, then fill with water only and weigh.
- For soils with soluble salts use kerosene instead of water.
- Gas jar method for coarse grained (> 4.75 mm) soils; Pycnometer for fine and medium grained soils.

Soil	Sand	Silt	Clay	Organic soil	Laterite	Gravel
G	2.65 - 2.67	2.67 - 2.70	2.70 - 2.80	1.4 - 2.3	2.75 - 3.0	2.65 - 2.68

- Use of G: needed for e, S, γ_d, hydrometer analysis, zero-air-void line and compaction.

8.3 In-situ (field) density

Determines γ and γ_d of soil in place; used for compaction control and bearing capacity.

$$\gamma = W / V \qquad \gamma_d = \gamma / (1 + w)$$

- Core cutter method (IS 2720 Part 29): for soft to medium cohesive soils (no gravel). Standard cutter: 100 mm dia, 130 mm height, volume about 981 cc. Drive the cutter, remove with soil, trim ends, weigh: $\gamma = (W_2 - W_1) / V_c$.
- Sand replacement method (IS 2720 Part 28): for gravelly and sandy soils. A hole of about 100 mm dia is dug, soil weighed (W), then the hole is filled with calibrated sand from a pouring cylinder. $V_{hole} = W_{sand\ in\ hole} / \gamma_{sand}$, where $W_{sand\ in\ hole} = W_1 - W_2 - W_{cone}$. Then $\gamma = W / V_{hole}$.
- Water displacement / wax method: for irregular lumps, soil coated with paraffin wax and weighed in air and water.
- Nuclear density gauge / rubber balloon method: quick and non-destructive.

Solved example 3 (sand replacement)

Excavated soil = 1.85 kg, water content 10 %. Sand filling the hole = 1.30 kg, density of sand = 1.45 g/cc. Find γ and γ_d.

- $V_{hole} = 1300/1.45 = 896.6$ cc; $\gamma = 1850/896.6 = 2.063$ g/cc
- $\gamma_d = 2.063/1.10 = 1.876$ g/cc

9. Grain Size Analysis (Mechanical Analysis)

Determines the relative proportion of different grain sizes in a soil, plotted as the particle size distribution curve.

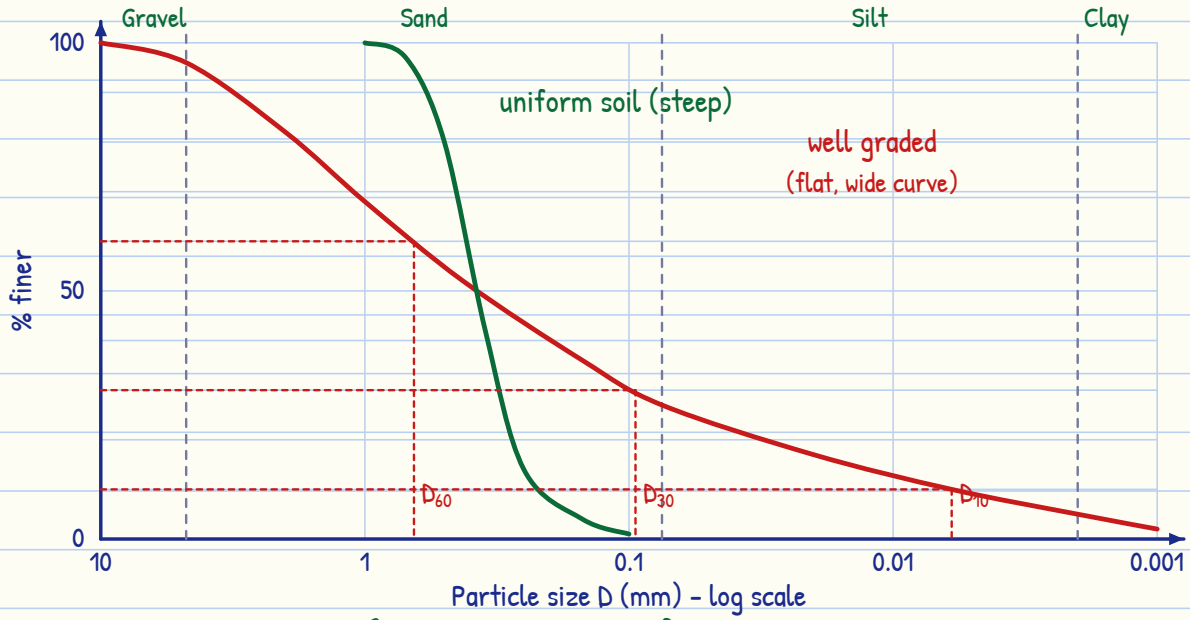
Soil fraction	Gravel	Coarse sand	Medium sand	Fine sand	Silt	Clay
Size (IS)	80 - 4.75 mm	4.75 - 2.0 mm	2.0 - 0.425 mm	0.425 - 0.075 mm	0.075 - 0.002 mm	< 0.002 mm

9.1 Sieve analysis (for particles > 75 micron)

- Used for coarse-grained soils. A set of IS sieves is arranged in descending order of opening: 4.75 mm, 2.0 mm, 1.0 mm, 600 micron, 425 micron, 300 micron, 212 micron, 150 micron, 75 micron, then the pan.
- Oven-dry soil (500 g - 1 kg) is washed over the 75 micron sieve (wet sieving for soils with fines), dried and then shaken for 10 minutes in a mechanical sieve shaker.
- Weight retained on each sieve is found.

% retained = (W retained on sieve / W total) x 100; % finer N = 100 - cumulative % retained

- Plot % finer (arithmetic) vs particle size D (log scale).



Grading parameters (from the curve)

- Effective size D_{10}: size such that 10 % of soil is finer. D_{30} and D_{60} similarly.

$$C_u = D_{60} / D_{10} \qquad C_c = (D_{30})^2 / (D_{10} \times D_{60})$$

- Well graded: gravel $C_u > 4$, sand $C_u > 6$ AND $1 < C_c < 3$ (all sizes present, good compaction).
- Poorly graded (uniform): does not satisfy the above; gap graded has some sizes missing (hump in curve).
- Steep curve means uniform soil, flat curve means well graded.

Example

D_{10} = 0.10 mm, D_{30} = 0.30 mm, D_{60} = 0.80 mm: C_u = 0.80/0.10 = 8; C_c = $0.30^2/(0.10 \times 0.80)$ = 1.125. $C_u > 6$ and C_c between 1 and 3, so it is a well graded sand.

9.2 Hydrometer analysis (for particles < 75 micron)

- Based on Stokes' law: the terminal velocity of a spherical particle settling in a liquid is $v = (\gamma_s - \gamma_w) D^2 / (18 \eta)$.

$$D = \text{sqrt}\left[18 \eta / (\gamma_s - \gamma_w) \times (H_e / t) \right] \quad (D \text{ in mm, } t \text{ in min})$$

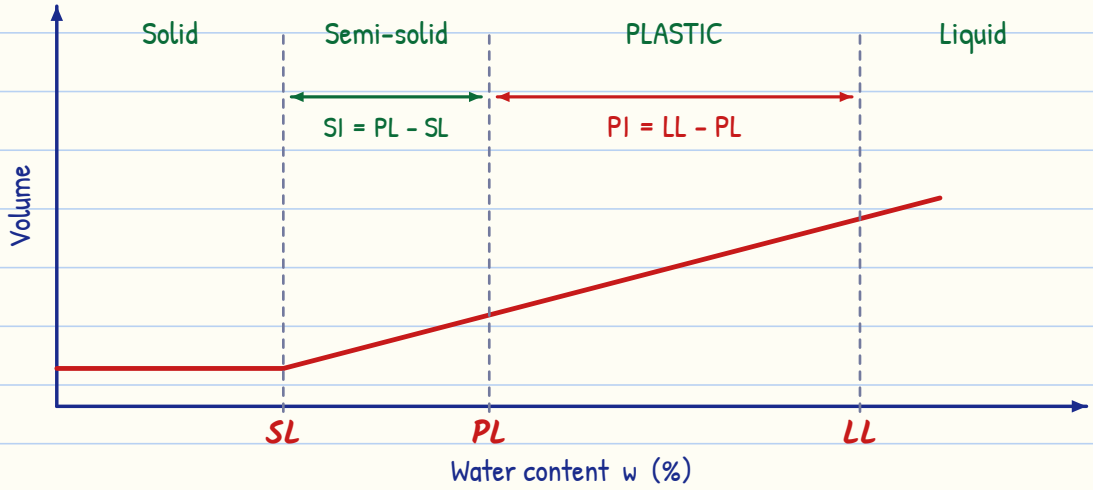
- η = viscosity of water, H_e = effective depth (from the surface to the centre of the hydrometer bulb), t = time of settling.
- Soil (50 g) + dispersing agent (sodium hexametaphosphate or sodium silicate, 2 %) is made into a 1000 ml suspension. Hydrometer readings (152H) are taken at 0.5, 1, 2, 4, 8, 15, 30, 60, 250 and 1440 minutes.
- Corrections: meniscus correction (C_m, always added), dispersing agent correction (C_d, subtracted) and temperature correction (C_t, + or -).

- % finer $N = (R_c \times G) / [\, W_s (G - 1)\,]$, where R_c is the corrected hydrometer reading.
- The results are combined with the sieve analysis result to obtain the complete grain size curve.
- Pipette method is more accurate but is a laboratory method; samples are drawn from a fixed depth with a pipette.

Stokes' law assumptions: particles are spherical, smooth, settle independently with no interference, and the velocity is small (laminar). Valid for 0.2 mm > D > 0.2 micron only.

10. Consistency Limits (Atterberg Limits)

A fine-grained soil can exist in four states depending on water content: solid, semi-solid, plastic and liquid. The boundary water contents were suggested by Atterberg (1911) and standardised by Casagrande.



Liquid limit w_L (LL)

Water content at which the soil changes from plastic to liquid state; it has a very small shear strength (about 2.7 kPa).

- Casagrande's apparatus (IS 2720 Part 5): soil paste (passing 425 micron) is placed in the brass cup, a groove is cut with a standard grooving tool and the cup is dropped through 1 cm at 2 drops/sec. LL = water content at which the groove closes by 12 mm in 25 blows.
- Take 3 - 4 trials with number of blows from 10 to 40 and plot flow curve (w vs log N). Flow index $I_f = (w_1 - w_2) / \log (N_2/N_1)$ = slope of flow curve.

- Cone penetration method: 30 degree cone weighing 148 g is released on the soil in a cup; LL is the water content corresponding to 25 mm penetration in 5 seconds.

Plastic limit w_P (PL)

Water content at which soil changes from semi-solid to plastic state. The soil is rolled by hand on a glass plate into a thread; PL = the water content at which the thread of 3 mm diameter just begins to crumble.

Shrinkage limit w_S (SL)

Water content below which there is no further decrease in volume of the soil on drying.

$$w_S = w_1 - [\,(V_1 - V_2) / W_d\,]\,\gamma_w \times 100$$

- w_1 = initial water content, V_1 = initial volume of wet pat, V_2 = volume of dry pat (by mercury displacement), W_d = dry weight.

Consistency indices

Index	Formula	Meaning
Plasticity index PI	$w_L - w_P$	range of water content over which soil is plastic
Shrinkage index SI	$w_P - w_S$	range of semi-solid state
Liquidity index LI	$(w - w_P) / PI$	LI < 0 stiff, 0 – 1 plastic, > 1 liquid
Consistency index CI	$(w_L - w) / PI$	CI = 1 – LI
Flow index I_f	slope of flow curve	rate of change of w with log N
Toughness index I_t	PI / I_f	shear strength at plastic limit
Activity A	PI / % clay (< 2 micron)	swelling potential

- Sensitivity S_t = (unconfined strength undisturbed) / (strength remoulded).
- Degree of plasticity (IS): PI 0 non-plastic, 1 – 7 low, 7 – 17 medium, 17 – 35 high, > 35 very high.

Solved example 4

LL = 48 %, PL = 25 %, natural water content = 36 %. Find PI, LI, CI and state of soil.

- PI = 48 – 25 = 23 %
- LI = (36 – 25)/23 = 0.478; CI = (48 – 36)/23 = 0.522
- 0 < LI < 1, so the soil is in the plastic state (medium stiff).

Sand and silt: PI = 0 (non-plastic). Higher PI means more clay content and higher compressibility and swelling.

II. I.S. Classification of Soils (IS 1498 - 1970)

Soils are grouped by particle size (coarse) and plasticity (fines). Three main divisions:

Division	Criterion	Groups
Coarse-grained	more than 50 % retained on 75 micron IS sieve	Gravel (G) and Sand (S)
Fine-grained	more than 50 % passes 75 micron IS sieve	Silt (M), Clay (C), Organic (O)
Highly organic	peat and other fibrous soils	Pt

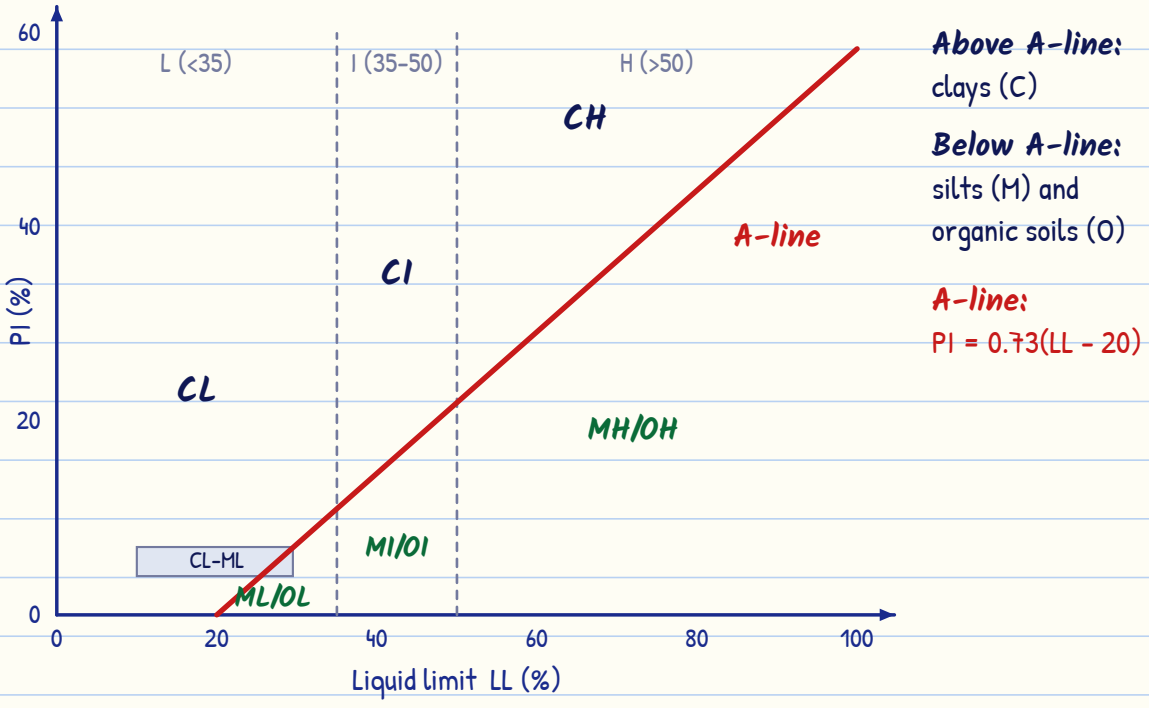
Coarse-grained soils

- Gravel (G): more than 50 % of the coarse fraction is retained on the 4.75 mm sieve. Sand (S): more than 50 % of coarse fraction passes 4.75 mm.
- Fines (passing 75 micron) < 5 %: W (well graded) or P (poorly graded) -> GW, GP, SW, SP.
- Fines > 12 %: M or C according to plasticity -> GM, GC, SM, SC.
- Fines 5 - 12 %: dual symbols e.g. GW-GM, SP-SC.

Group	GW	GP	SW	SP
Condition	$C_u > 4$ and $1 < C_c < 3$	not GW	$C_u > 6$ and $1 < C_c < 3$	not SW

Fine-grained soils (use the plasticity chart)

- LL < 35 %: L (low compressibility), 35 - 50 %: I (intermediate), LL > 50 %: H (high).
- Above the A-line (PI = 0.73 (LL - 20)): inorganic clays CL, CI, CH.
- Below the A-line: inorganic silts ML, MI, MH and organic silts / clays OL, OI, OH.
- CL-ML: hatched zone (PI between 4 and 7) near the A-line; also called borderline soils.



Group symbols – summary

Symbol	Meaning	Symbol	Meaning
GW	well graded gravel	GM	silty gravel
GP	poorly graded gravel	GC	clayey gravel
SW	well graded sand	SM	silty sand
SP	poorly graded sand	SC	clayey sand
ML	silt of low compress.	CL	clay of low plasticity
MI	silt, intermediate	CI	clay, intermediate
MH	silt of high compress.	CH	clay of high plasticity
OL/OI/OH	organic silts and clays	Pt	peat

Field identification of soils (quick tests)

- Dry strength, dilatancy (shaking test), toughness (thread test) and visual inspection (colour, odour) classify fine soils in the field.
- Silts show quick dilatancy and low dry strength; clays show no dilatancy and high dry strength; organic soils have a spongy feel and odour.

12. Quick Revision – Important Formulas

Quantity	Formula
Void ratio / porosity	$e = V_v/V_s$; $n = e/(1+e)$
S e = w G	Dry unit weight $\gamma_d = G\gamma_w/(1+e) = \gamma/(1+w)$
Saturated unit weight	$\gamma_{sat} = (G + e)\gamma_w/(1+e)$
Submerged unit weight	$\gamma' = (G - 1)\gamma_w/(1+e)$
Relative density	$D_r = (e_{max} - e)/(e_{max} - e_{min})$
Uniformity coefficient	$C_u = D_{60}/D_{10}$
Coefficient of curvature	$C_c = D_{30}^2/(D_{10} D_{60})$
Plasticity index	PI = LL – PL
Liquidity index	LI = (w – PL)/PI
A-line	PI = 0.73 (LL – 20)

Important questions (exam)

- Explain the formation of soils and the different types of soil structures.
- Define void ratio, porosity, degree of saturation and derive e, S, w, G relation: S e = w G.

- Derive γ_{sat}, γ_d and γ' in terms of G and e.
- Explain the sand replacement method for finding in-situ density.
- Describe the hydrometer analysis with Stokes' law.
- Explain Atterberg limits and the indices of consistency; how are they determined?
- Explain the IS classification of soil and the plasticity chart.